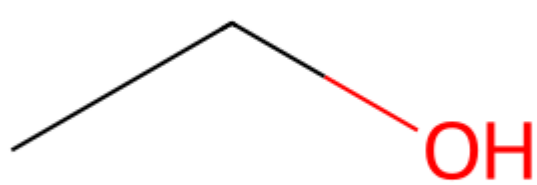


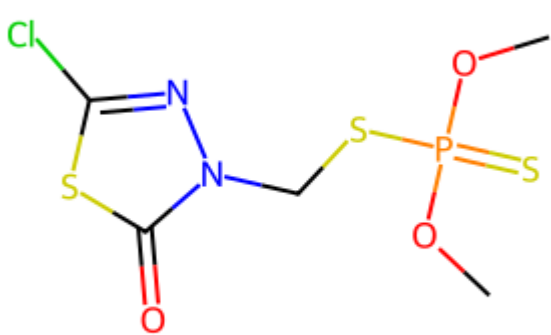


{'analysis_time': 0.0,
'first_solution_time': 61.638841,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 18.227535687182588,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 18.227535687182588,
'retro_star_value': 18.227535687182588}

Reaction
{'analysis_time': 0.142126,
'first_solution_time': 61.638841,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 18.227535687182588,
'retro_star_reaction_number': 18.227535687182588,
'retro_star_rxn_cost': 6.950317128070204,
'retro_star_value': 18.227535687182588}



{'analysis_time': 0.142126,
'first_solution_time': 6.01104,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.44870909519674496,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 0.44870909519674496,
'retro_star_value': 18.227535687182588}



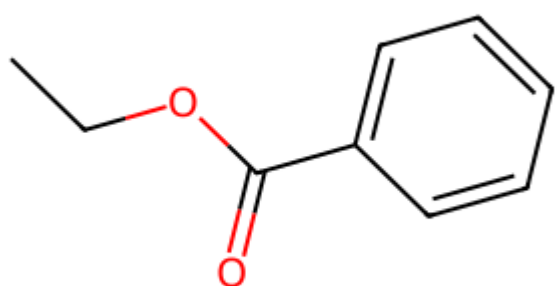
{'analysis_time': 0.142126,
'first_solution_time': 61.638841,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 10.828509463915639,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 10.828509463915639,
'retro_star_value': 18.227535687182588}



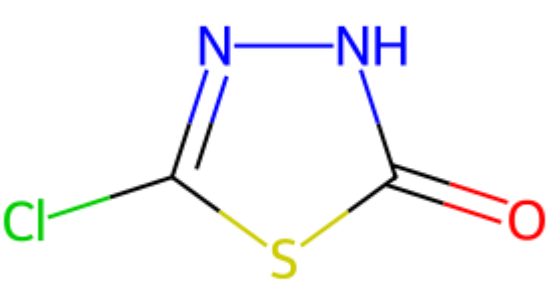
Reaction
{'analysis_time': 6.01104,
'first_solution_time': 6.01104,
'num_calls_rxn_model': 43,
'num_calls_value_function': 1133,
'retro_star_min_cost': 8.19745595043746,
'retro_star_reaction_number': 8.19745595043746,
'retro_star_rxn_cost': 8.19745595043746,
'retro_star_value': 25.976282542423306}



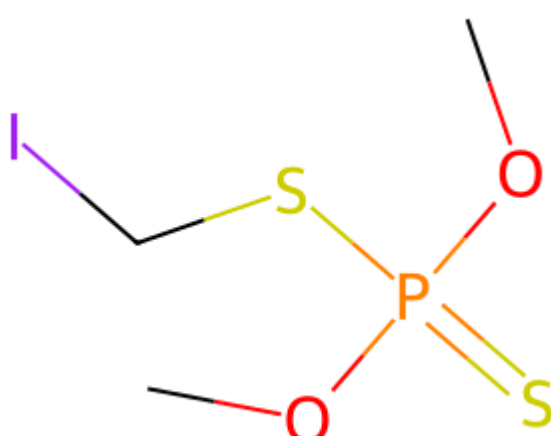
Reaction
{'analysis_time': 6.537774,
'first_solution_time': 61.638841,
'num_calls_rxn_model': 47,
'num_calls_value_function': 1262,
'retro_star_min_cost': 10.828509463915639,
'retro_star_reaction_number': 10.828509463915639,
'retro_star_rxn_cost': 9.482955789591976,
'retro_star_value': 18.227535687182588}



{'analysis_time': 6.01104,
'first_solution_time': 6.01104,
'num_calls_rxn_model': 43,
'num_calls_value_function': 1133,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 25.976282542423306}

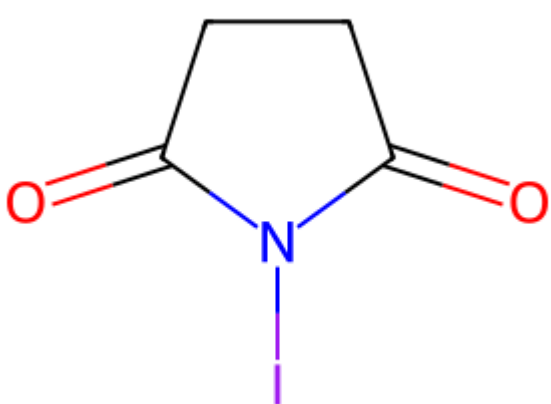


{'analysis_time': 6.537774,
'first_solution_time': 6.537774,
'num_calls_rxn_model': 47,
'num_calls_value_function': 1262,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 18.227535687182588}

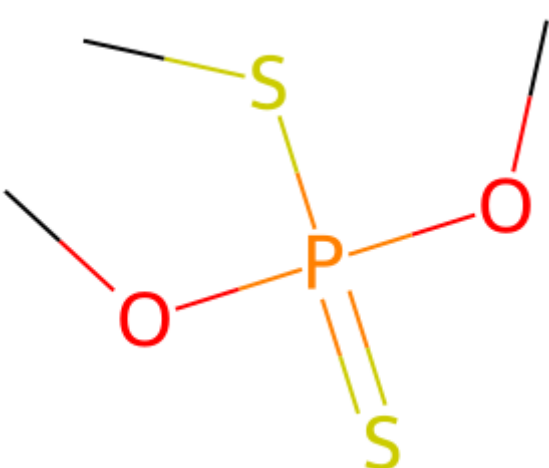


{'analysis_time': 6.537774,
'first_solution_time': 61.638841,
'num_calls_rxn_model': 47,
'num_calls_value_function': 1262,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 1.345553674323663,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 1.345553674323663,
'retro_star_value': 18.227535687182588}

Reaction
{'analysis_time': 61.638841,
'first_solution_time': 61.638841,
'num_calls_rxn_model': 378,
'num_calls_value_function': 20935,
'retro_star_min_cost': 1.345553674323663,
'retro_star_reaction_number': 1.345553674323663,
'retro_star_rxn_cost': 1.345553674323663,
'retro_star_value': 18.227535687182588}



{'analysis_time': 61.638841,
'first_solution_time': 61.638841,
'num_calls_rxn_model': 378,
'num_calls_value_function': 20935,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 18.22753568718259}



{'analysis_time': 61.638841,
'first_solution_time': 61.638841,
'num_calls_rxn_model': 378,
'num_calls_value_function': 20935,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 18.22753568718259}